

## Key features of parabolas



- 1
  - a Downwards
  - b Upwards
- 2 Upward
- 3 No, because the entire parabola will be above the  $x$ -axis. Also, there is no solution for the equation  $x^2 + 6 = 0$ .
- 4
  - a No
  - b Yes
  - c No
  - d Yes
- 5
  - a  $x = 0, 4$
  - b  $y = 0$
  - c 4
- 6
  - a Concave up
  - b  $y = 9$
  - c  $y = 1$
  - d  $x = -2$
  - e  $x < -2$
- 7
  - a  $(-2, 0)$
  - b Vertex:  $(-2, 0)$
  - c
    - i False
    - ii True
    - iii True
    - iv True
- 8
  - a 2
  - b 1
- 9 Concave up
- 10
  - a  $(-4, 2)$
  - b 2
- 11  $x = 3$
- 12
  - a No
  - b No
  - c Yes
- 13 Quadratic function  $A$



15  $y = 25$

16

a  $x = 14, 6$

b  $x = 10$

c Concave down

d  $y = 16$

17 The  $x$ -coordinate of the vertex of the parabola occurs at  $x = \frac{-b}{2a}$ . The  $y$ -coordinate of the vertex is found by substituting  $x = \frac{-b}{2a}$  into the parabola's equation and evaluating the function at this value of  $x$ .

18  $x = \frac{1}{4}$

19  $x = -3, 2$

20

a False

b True

21  $x = \frac{-3 \pm \sqrt{93}}{6}$

## Graphing quadratics

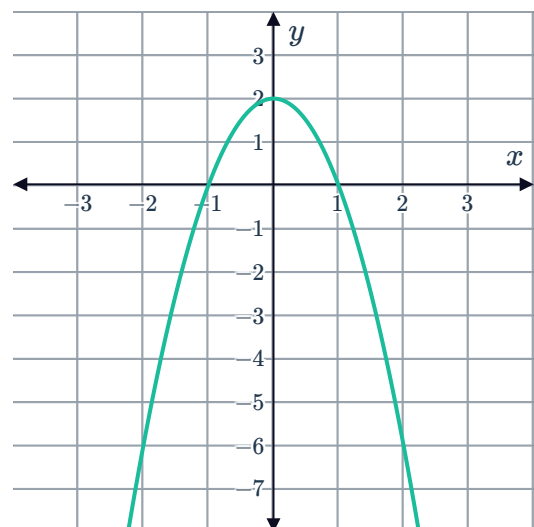
22

a Concave down

c  $(0, 2)$

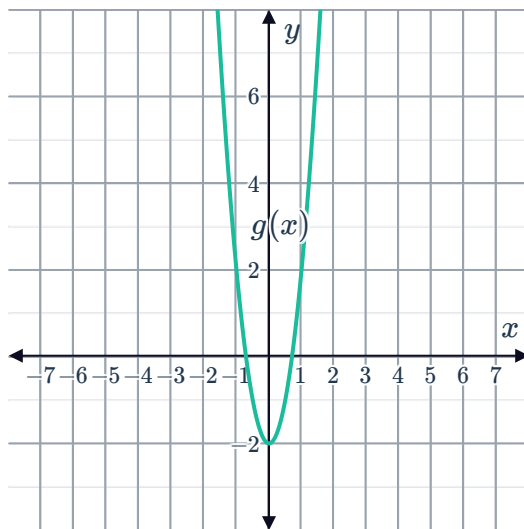
b More steep

d

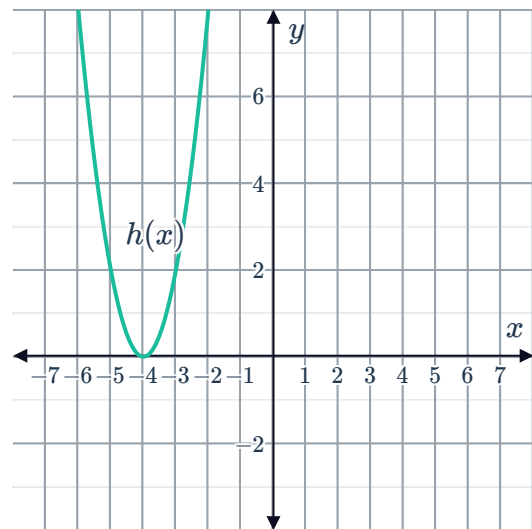


23

a



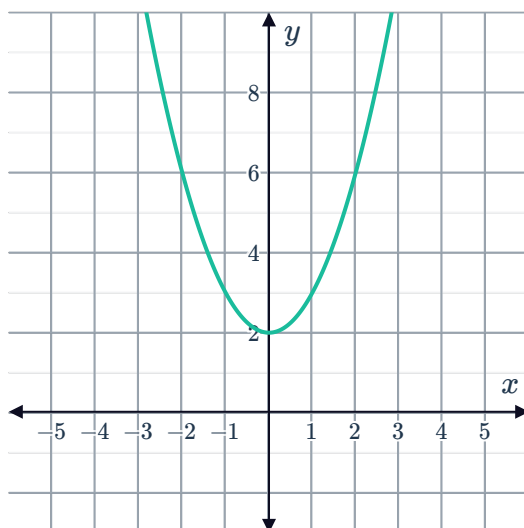
b



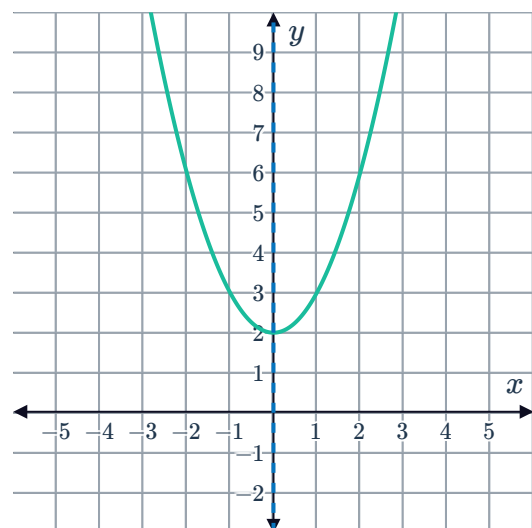
24  $y = -x^2 + 5$

25

a



b



c  $(0, 2)$

26

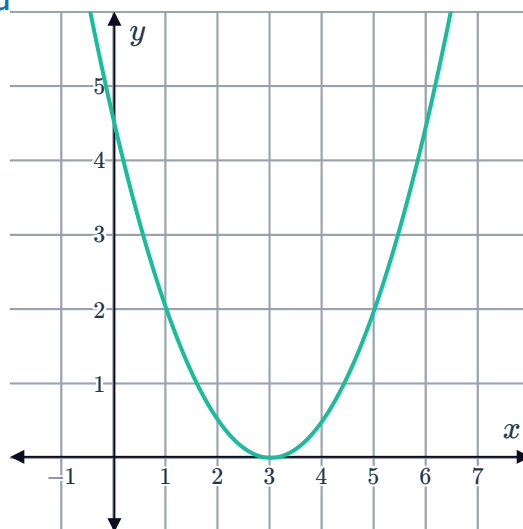
a Concave up

b Less steep

c  $(3, 0)$



d



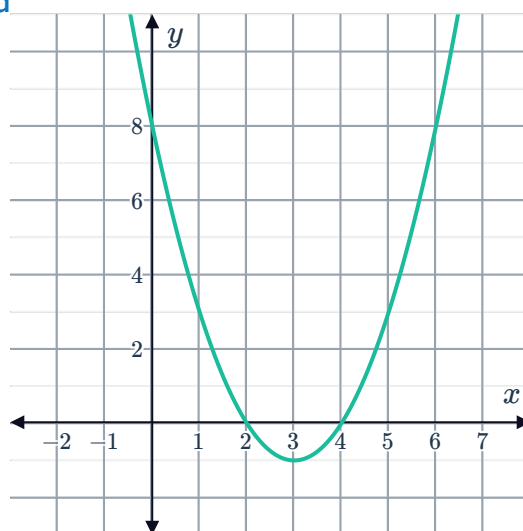
27

a  $x = 4, 2$

b  $y = 8$

c  $(3, -1)$

d



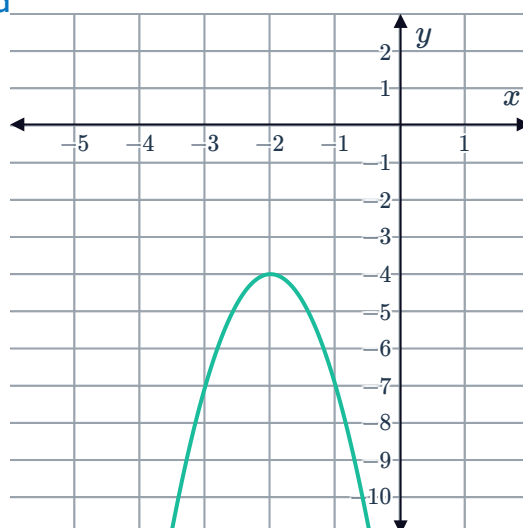
28

a  $(-2, -4)$

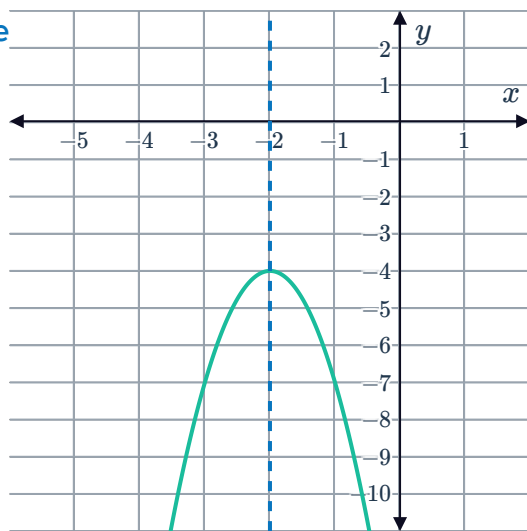
b  $x = -2$

c  $y = -7$

d

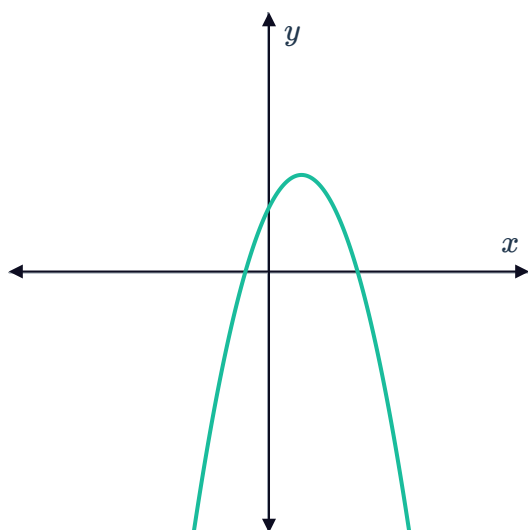


e

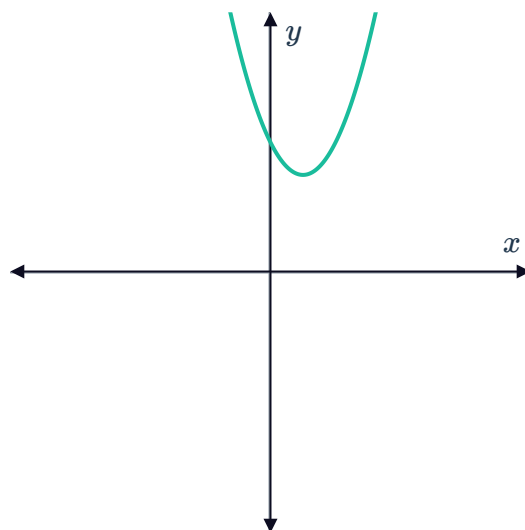


29

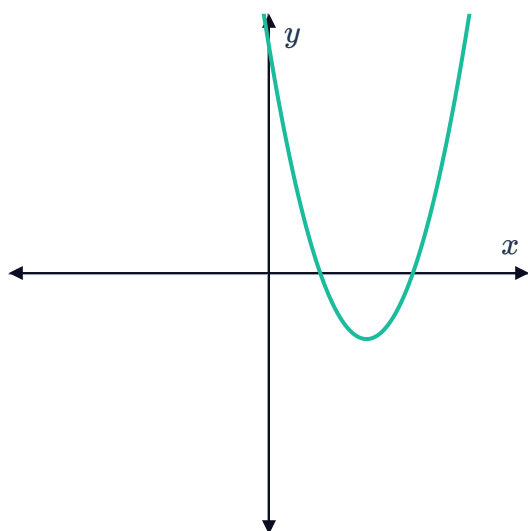
a



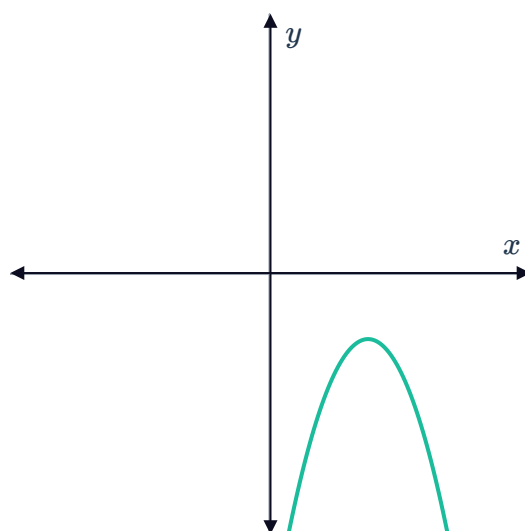
b



c



d



30

a 8

c

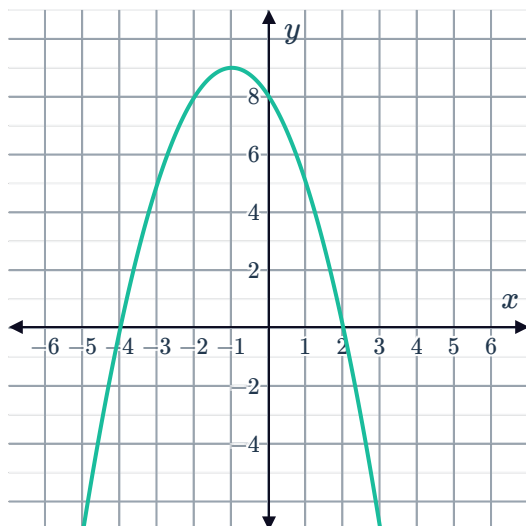
|     |    |    |    |   |    |
|-----|----|----|----|---|----|
| $x$ | -5 | -3 | -1 | 1 | 3  |
| $y$ | -7 | 5  | 9  | 5 | -7 |



b  $x = 2, -4$

d  $(-1, 9)$

e



31

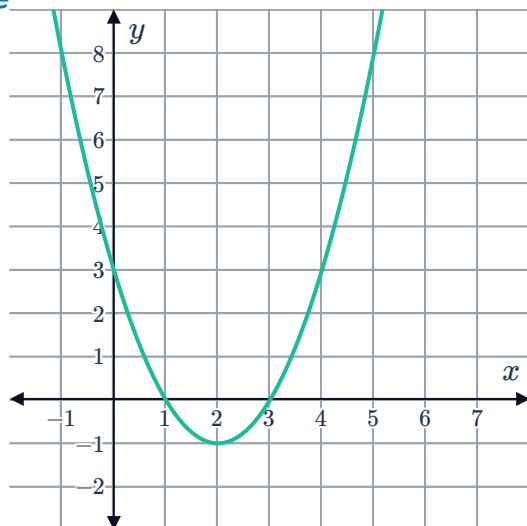
a  $y = 3$

b  $x = 3, 1$

c  $x = 2$

d  $(2, -1)$

e



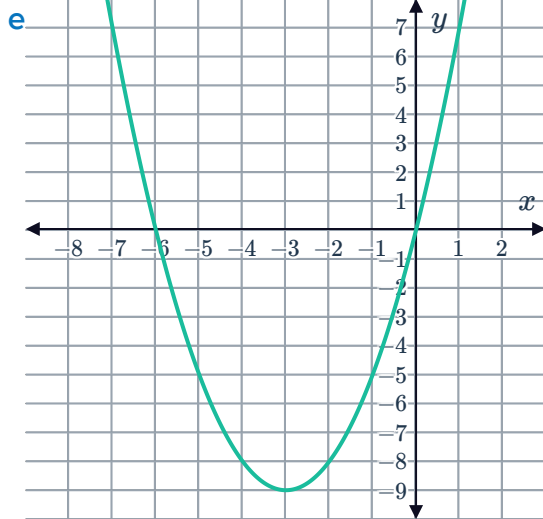
32

a  $y = 0$

b  $x = 0, -6$

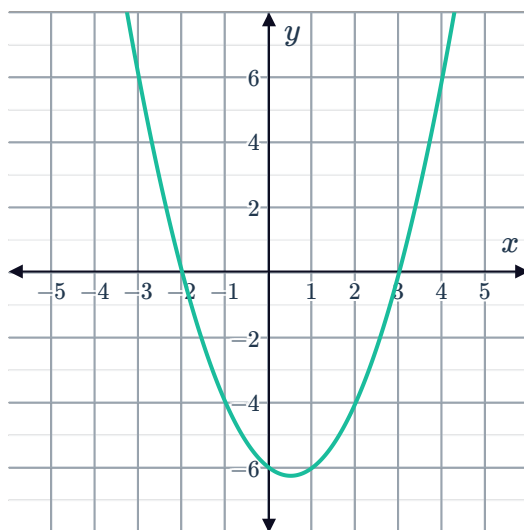
c  $x = -3$

d  $(-3, -9)$

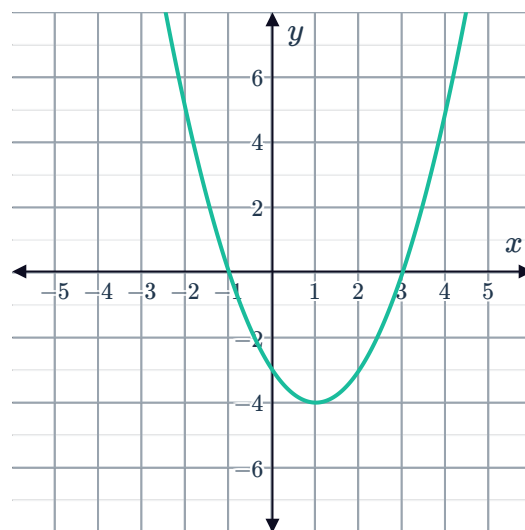


33

a

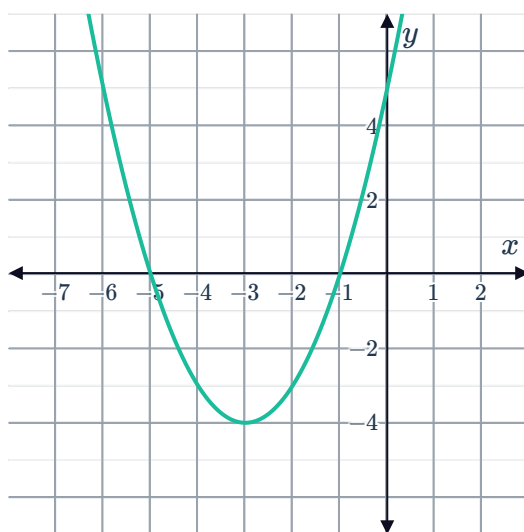


b

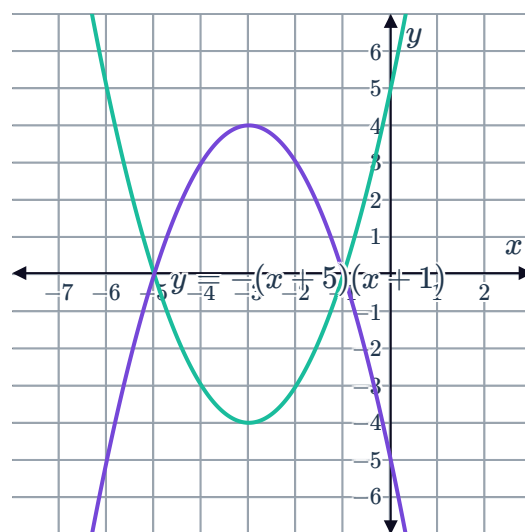


34

a



b



35

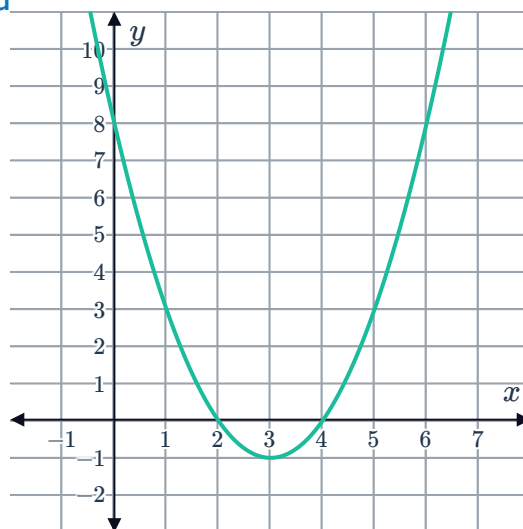
a  $(x - 2)(x - 4)$

b  $x = 2, 4$

c  $(3, -1)$



d



36

a  $y = (x + 4)(x - 3)$

c  $y = -12$

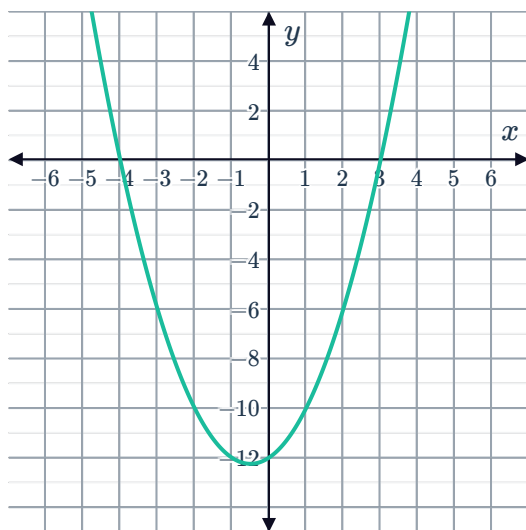
e  $\left(-\frac{1}{2}, -\frac{49}{4}\right)$

b  $x = -4, 3$

d  $x = -\frac{1}{2}$

f Concave up

g



37

a  $y = -1$

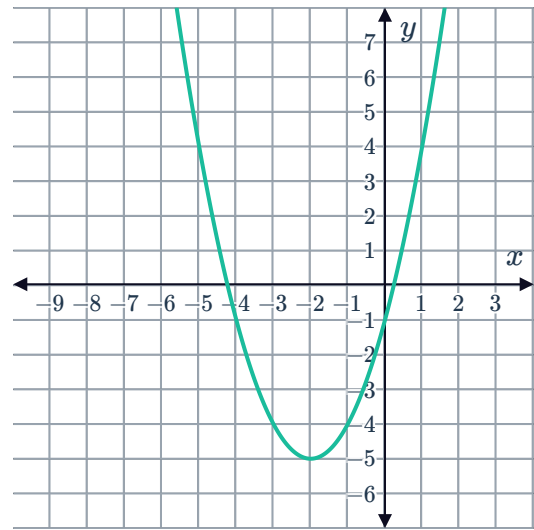
b  $(-2, -5)$



c Concave up



d

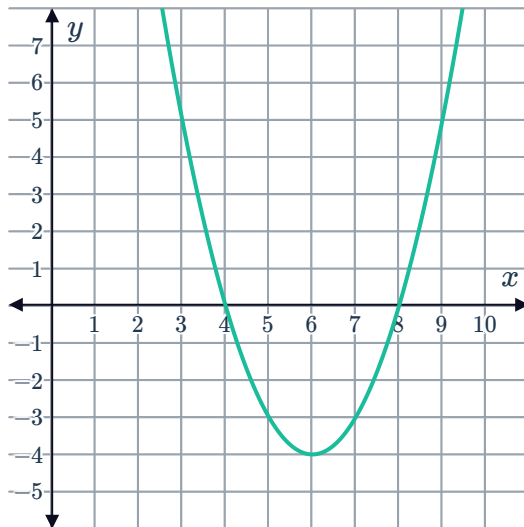


38

a  $x = 8, 4$

b  $(6, -4)$

c

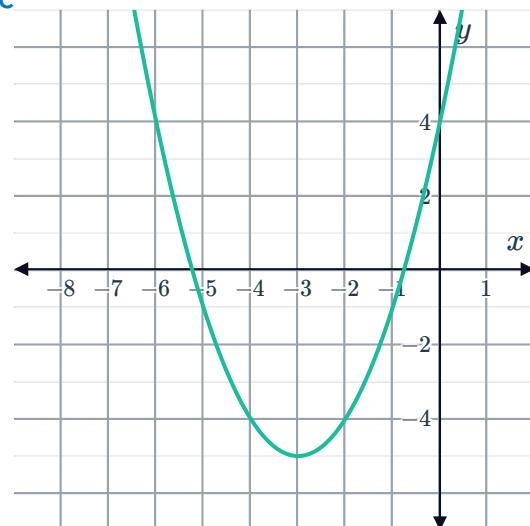


39

a  $x = -3$

b  $-5$

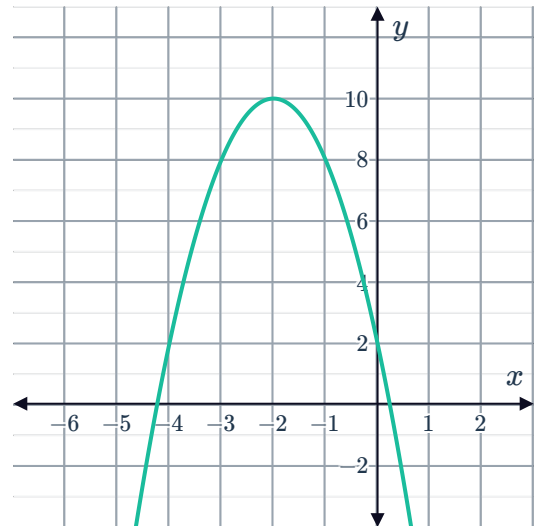
c



40 a  $(-2, 10)$



b

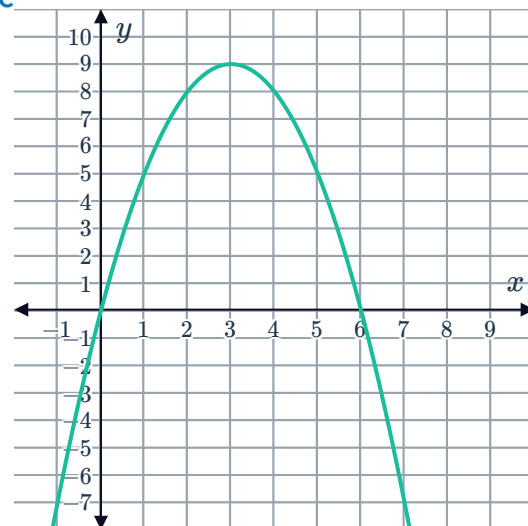


41

a  $x = 0,6$

b  $(3, 9)$

c



42

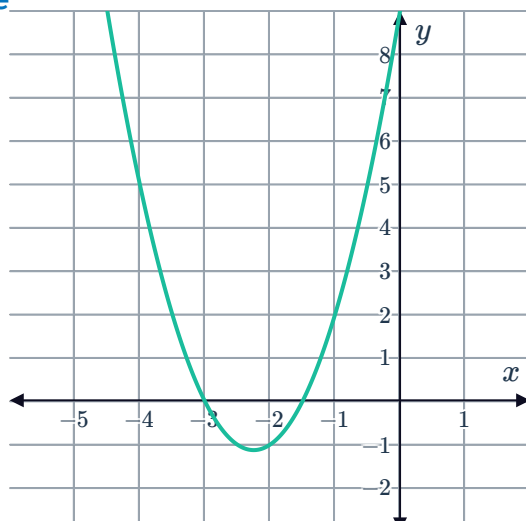
a  $x = -\frac{3}{2}, -3$

b  $y = 9$

c  $x = -\frac{9}{4}$

d  $y = -\frac{9}{8}$

e





43

a

i  $(8, 7)$

ii  $(0, 263)$

b

i  $(-6, 4)$

ii  $(0, -140)$

44

a

$(0, -12)$

b No

c  $x < 0$

45

a

$x = -3.1$

b

$-16.61$

46

a

$x = 2, 6$

b

$y = -24$

c

$x = 4$

d

$y = 8$

47

a

$x = 1.55$

b

$y = 2.95$

c

Yes

d

$x = 3.58, -0.47$

48

a

$x = 2.75$

b

$y = -9.10$

c

Yes

d

$x = -0.42, 5.91$